

DIAMOND – CALCIUM NITRATE
TETRAHYDRATE

Ca(NO₃)₂ · 4H₂O | Ultra-Processed | Quadruple-Filtered | Virgin Raw Materials
Available in Solid Crystalline and Aqueous Solution forms

PRODUCT DESCRIPTION

Diamond Calcium Nitrate Tetrahydrate [Ca(NO₃)₂·4H₂O, CAS 13477-34-4] is an ultra-processed, highly purified inorganic salt manufactured by Sterling Chemicals using only virgin raw materials and a proprietary quadruple-filtration process. The result is a product with consistently low levels of chloride, sulfate, and iron — critical quality parameters in demanding applications such as latex coagulation and technical-grade synthesis.

Diamond grade is available in two commercial forms — **Solid Crystalline** and **Aqueous Solution** — enabling customers to select the format best suited to their process. Manufactured under ISO 9001:2015.

≥ 99.0% Purity (Tetrahydrate)	≥ 11.5% (as N) Nitrogen Content	≥ 16.5% (as Ca) Calcium Content
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PRODUCT SPECIFICATIONS

Parameter	Solid Crystalline	Aqueous Solution
Appearance	White crystals	Clear, colourless to pale-yellow liquid
Ca(NO ₃) ₂ content	≥99.0% (tetrahydrate basis)	48–52% w/w
Calcium (as Ca)	≥16.5%	≥8.3%
Nitrogen (as N)	≥11.5%	≥5.9%
Density	~1.90 g/cm ³ (bulk: ~0.95 g/cm ³)	1.45–1.65 g/cm ³
pH (5% aqueous)	5.0 – 7.0	5.0 – 7.0
Chloride (as Cl)	≤ 50 ppm	≤ 0.010%
Sulfate (as SO ₄)	≤ 50 ppm	≤ 0.020%
Iron (as Fe)	≤ 10 ppm	≤ 10 ppm
Water insolubles	≤0 .005%	—
Moisture (solid)	≤ 0.5%	—
Form factor	Crystalline	Ready-to-use liquid

* Aqueous solution concentration is nominally 50% w/w Ca(NO₃)₂. Custom concentrations available on request.

FORM CHARACTERISTICS

Solid Crystalline

White, odourless crystals. Highly water-soluble (dissolves readily in ambient water). Hygroscopic — store in sealed, moisture-resistant packaging.

Advantages:

- Extended shelf life and stability in storage
- Precise dosing by weight — ideal for batch processes
- Lower freight cost per unit of active content
- Flexible dissolution: dissolve to any target concentration on-site

Handling:

Dissolve in water at ambient temperature with moderate agitation. Use corrosion-resistant (SS 316 or HDPE) vessels. Wear gloves and eye protection during handling.

Aqueous Solution

Clear, colourless to pale-yellow liquid at ~50% w/w concentration. Ready-to-use without dissolution. Exhibits low viscosity and excellent pump-ability at ambient temperatures.

Advantages:

- Ready-to-use — eliminates on-site dissolution step
- Accurate metering with standard dosing pumps
- Faster process integration and startup
- Consistent concentration batch to batch

Handling:

Pump directly from IBC or drum using HDPE or SS 316 compatible dosing equipment. Avoid aluminium and zinc contact. Store above 10 °C to prevent crystallisation.

APPLICATIONS

Application	Industry	Notes
Latex Coagulation	Rubber & Latex Industry	Precipitates latex without discolouration; chloride-free grade prevents product defects. Preferred by 99% of Indian latex producers.
Concrete Acceleration	Construction & Infrastructure	Accelerates cement hydration at low temperatures; reduces setting time and improves early compressive strength.
Corrosion Inhibition	Concrete Reinforcement	Calcium nitrate in concrete mix passivates rebar, significantly extending the service life of reinforced structures.
Micronutrient Fertilisation	Horticulture & Agriculture	Dual-nutrient source: soluble calcium for cell wall integrity and fast-release nitrate-nitrogen for plant uptake.
Industrial Synthesis	Chemical Manufacturing	Precursor and reagent for nitrate salt synthesis, oxidising bath formulations, and heat-transfer media.

QUALITY & MANUFACTURING

- ISO 9001:2015 certified quality management system
- Quadruple filtration for ultra-low impurity levels
- Virgin raw materials — no recycled or low-grade input streams
- In-house QC with ICP-OES, titration, and gravimetric methods
- Batch COA available with every shipment
- Temperature-controlled storage before despatch
- Dedicated production line — no cross-contamination risk
- Traceability from raw material to finished goods

PACKAGING & LOGISTICS

Form	Small Pack	Intermediate Pack	Bulk
Solid Crystalline	25 kg HDPE bag	1000 kg jumbo bag	-
Aqueous Solution (50%)	200 L HDPE drum	1000 L IBC	Tanker

Shelf life: 24 months (solid crystalline, sealed); 12 months (aqueous solution). Custom packaging and labelling available for OEM customers.

STORAGE & SAFETY

Store in a cool, dry, well-ventilated area away from reducing agents, combustible materials, and organic substances. Calcium nitrate is an oxidiser. Full Safety Data Sheet (SDS) available on request.

COMMERCIAL TERMS

- Ex-works: Ankleshwar, Gujarat
 - Payment: 100% advance (TT / RTGS / NEFT)
 - Lead time: 7–10 working days (ex-stock: 3–5 days)
- All prices exclusive of 18% GST
 - Export: CIF or FOB Hazira / Nhava Sheva / Mundra
 - Sample orders available for qualified enquiries

To place an order or request a quotation, please contact:

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